Admin

- Assign 2
  Extension demos
  Retest on open issues available after test results returned
- Assign 3
  printf perseverance and pride!

Today: Thanks for the memory!

Loading, how an executable file becomes a running program

Address space layout, organization of memory

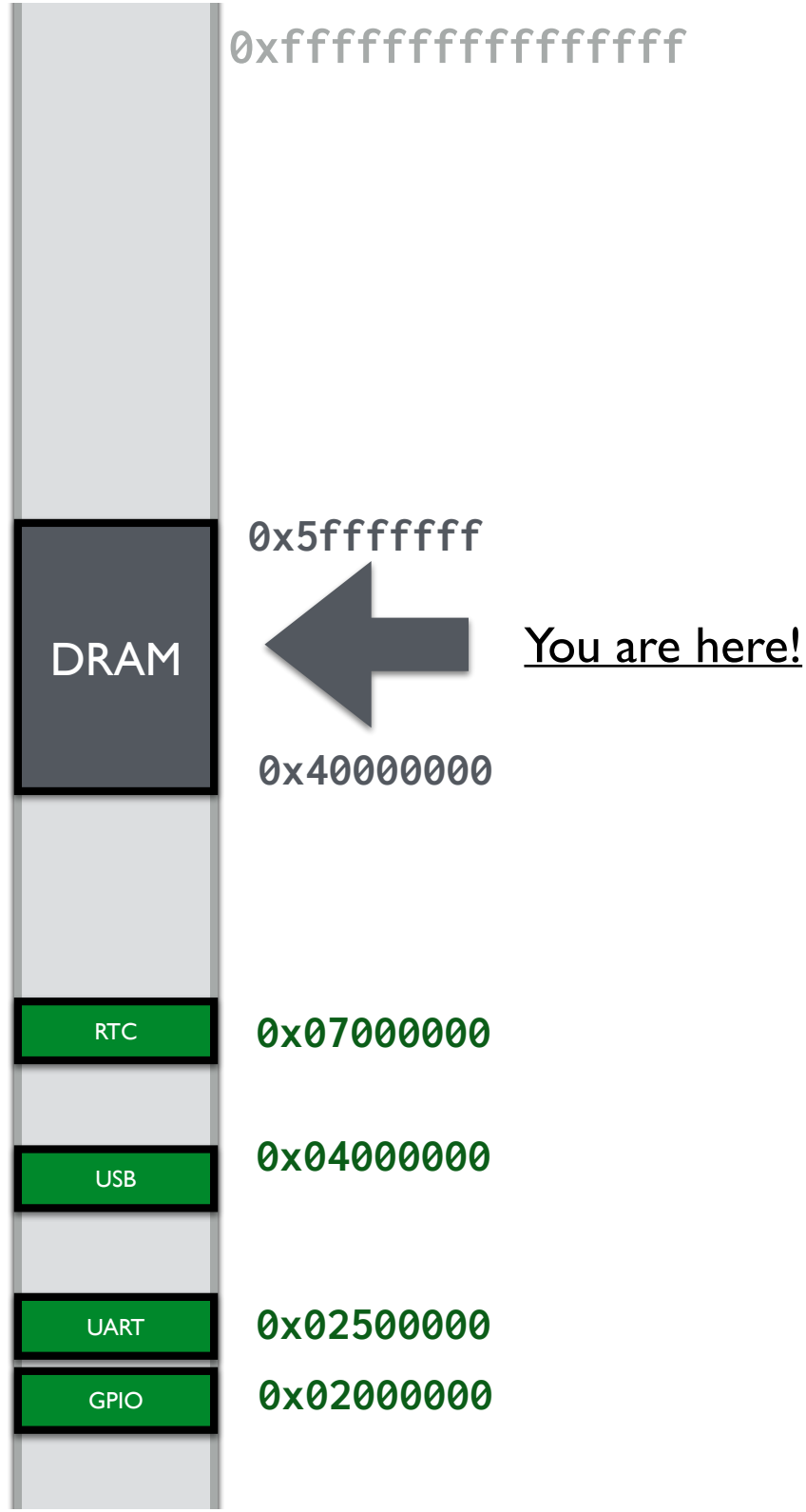
Runtime stack, stack frames, stack allocation

Heap allocation, malloc and free

Memory map

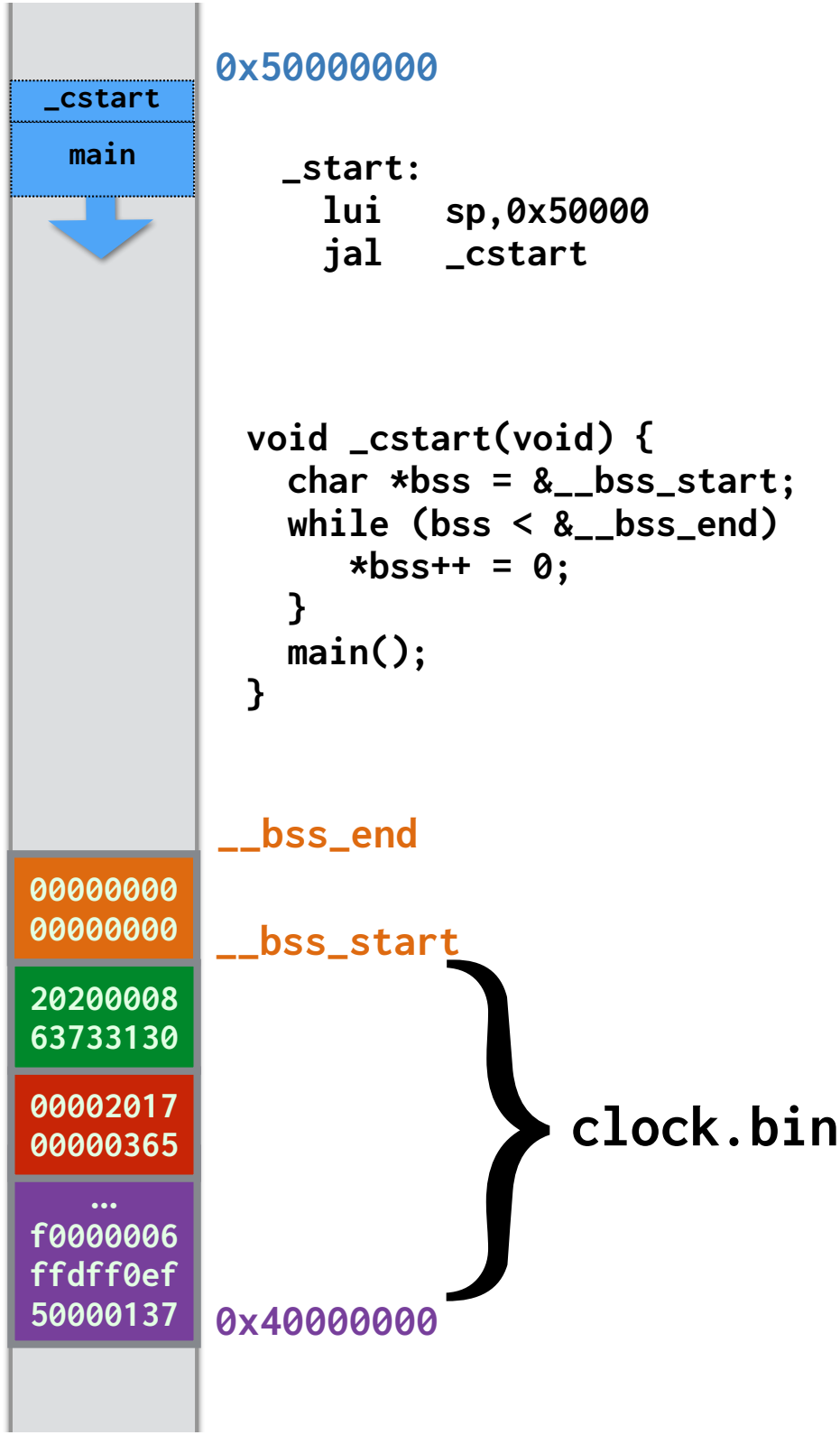
64-bit address space
`0x0 - 0xffffffffffffffff`

512MB physical RAM at
`0x40000000-0x5fffffff`



Ref: D1-H User Manual p.45

```
SECTIONS
{
  .text 0x40000000 :{ *(.text.start)
                      *(.text*)}
  .rodata :          { *(.rodata*) }
  .data :            { *(.data*) }
  __bss_start        = .;
  .bss :             { *(.bss*) }
  __bss_end          = . ;
}
```



(zeroed data) .bss

(initialized data) .data

(read-only data) .rodata

.text

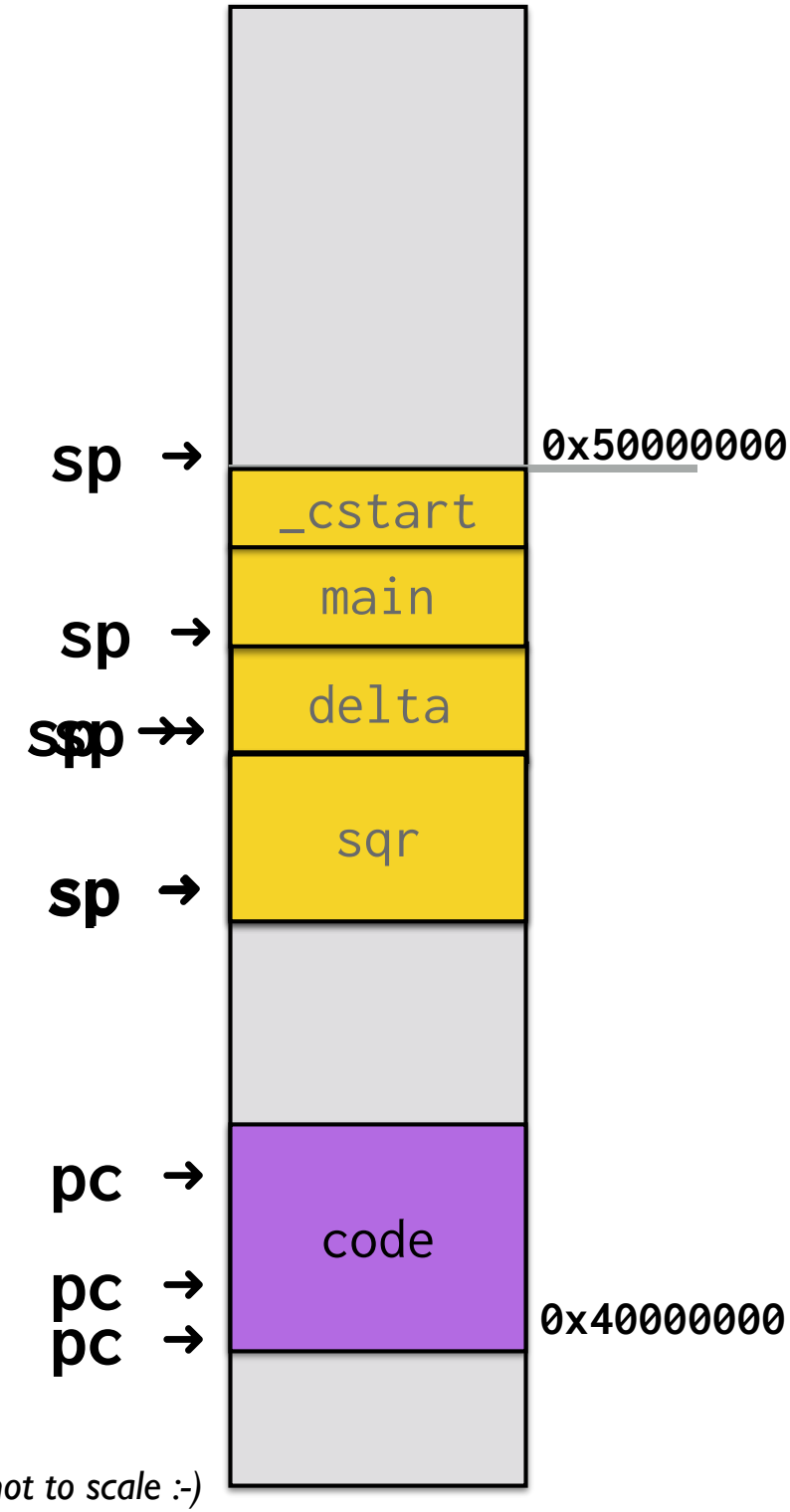
```
$ xfel write 0x40000000 clock.bin
$ xfel exec  0x40000000
```

```
// start.s
lui   sp,0x5000
jal   _cstart
```

```
void main(void)
{
   delta(3, 7);
}

int delta(int a, int b)
{
  int diff = sqr(a) - sqr(b);
  return diff;
}

int sqr(int v)
{
  return v * v;
}
```



Memory diagram not to scale :-)

Frame pointer

Designate register `s0` for use as **frame pointer** `fp`

Stack pointer `sp` is top of stack, additional register `fp` marks boundary between current stack frame and previous

Each frame saves previous `fp` — this gives reliable way to backtrace entire stack

CFLAGS to enable: `-f-no-omit-frame-pointer`

Add instructions to prolog/epilog that set up `fp` on entry and restore saved on exit

Tracing stack frames

Prolog

Adjust stack pointer to make space (16 bytes + rest)

Save registers `ra` and `fp` on stack, first two slots

Set `fp` to where `sp` was (end of prev stack frame)

Body

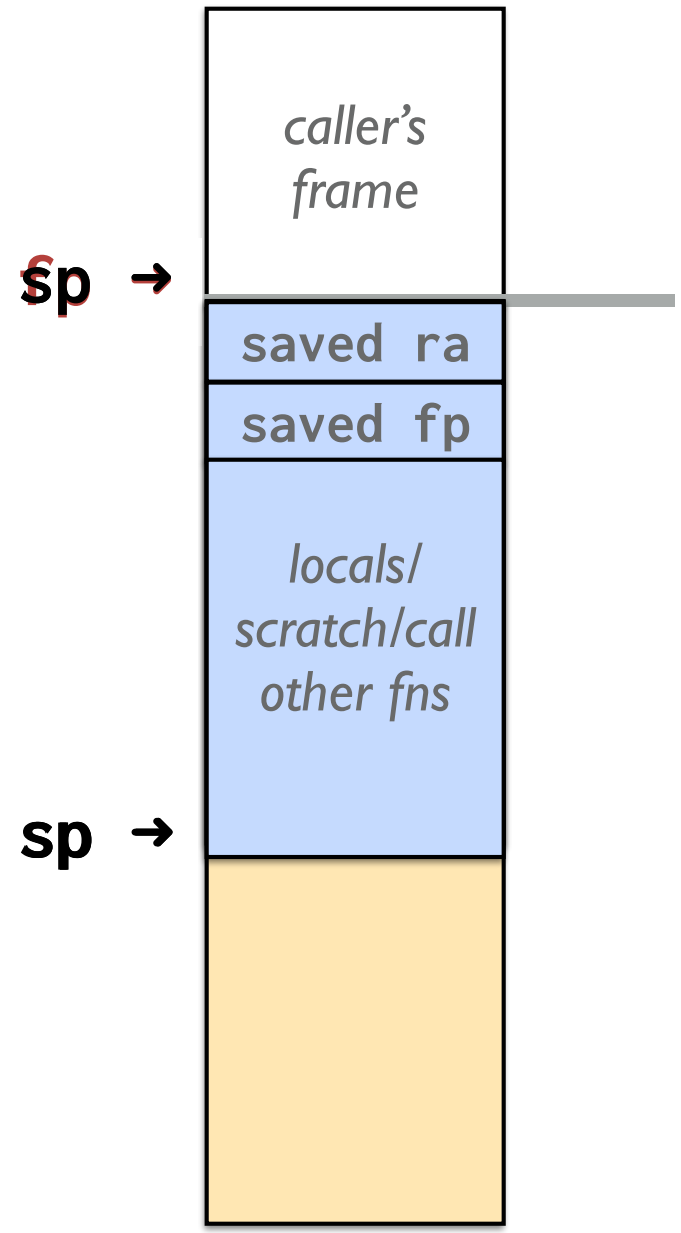
`sp` and `fp` anchors mark start/end of current frame

access data on stack `sp`-relative

Epilog

Restore `ra` and `fp` to saved values on stack

Adjust `sp` to remove frame

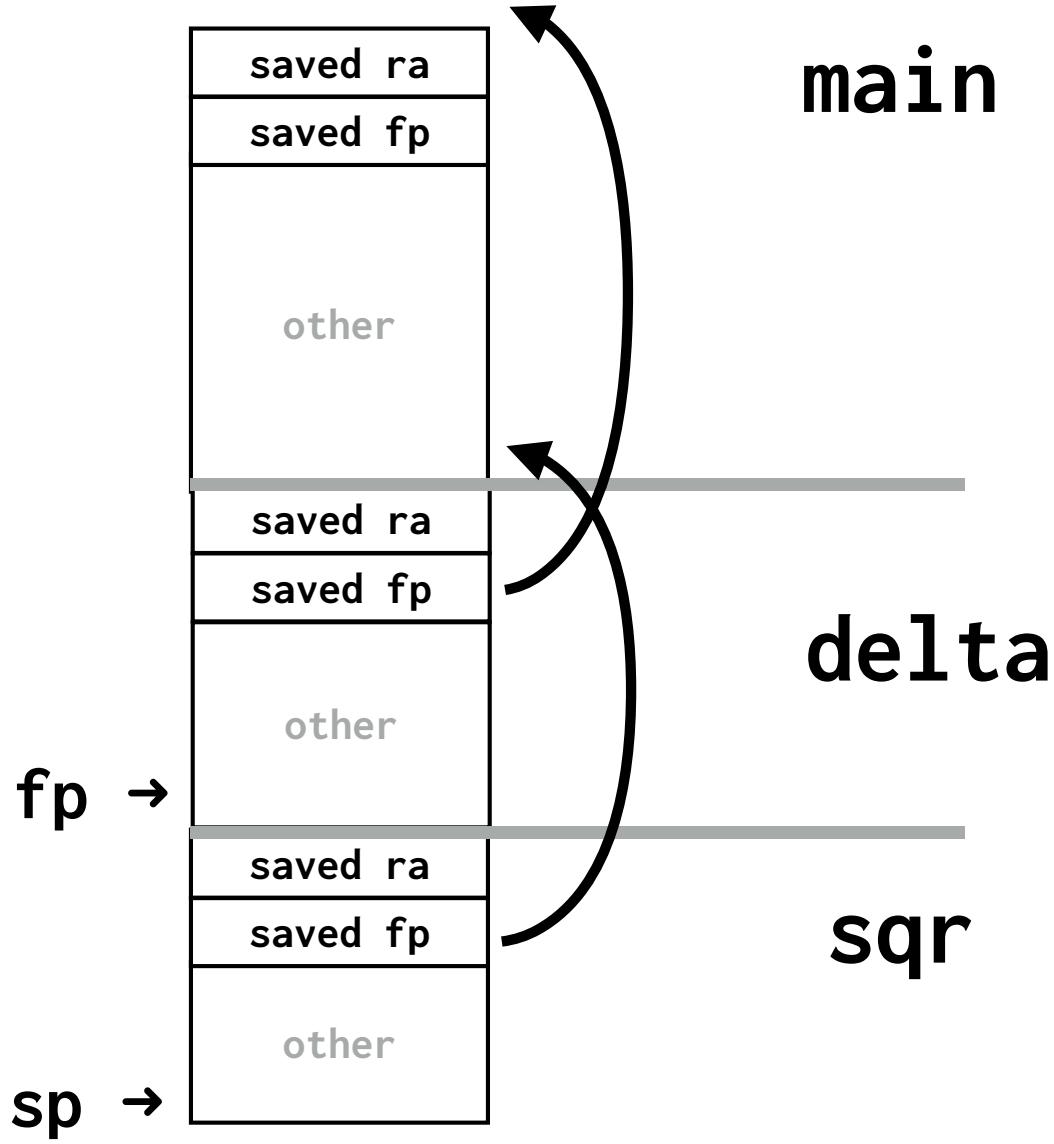


Linked chain of frame pointers

Can traverse from innermost frame (**sqr**) to frame of its caller (**delta**), from there to its caller (**main**) …

```
// start.s

// init fp = 0 as termination
li fp,0
lui sp,0x50000
jal _cstart
```



other =
more saved regs, locals, scratch

Deep dive into full frame coming up in lab 4!

Frame pointer tradeoffs

+ Anchored `fp`, offsets are constant
+ Standard frame layout enables runtime introspection
+ Backtrace for debugging, instrumentation

- `fp` register removed from general pool
- Added 3 instructions in prolog,epilog to manage `fp`
- Adds 8 bytes to frame size, another saved register

Aside: "Fedora's tempest in a stack frame"

https://developers.redhat.com/articles/2023/07/31/frame-pointers-untangling-unwinding

https://fedoraproject.org/wiki/Changes/fno-omit-frame-pointer https://lwn.net/Articles/919940/

We have global storage ...

+ **Convenient**

  Fixed location, shared across entire program

  No explicit allocate/deallocate

+ **Fairly efficient, plentiful**

  (But cost to send over serial line to bootloader)

+/- **Scope and lifetime is global**

  No encapsulation, hard to track use/dependencies

  One shared namespace, possibility of conflicts

  Heavily frowned upon stylistically

... and we have stack storage ...

+ **Convenient**

  Automatic alloc/dealloc on function entry/exit

+ **Efficient, fairly plentiful**

  (But finite size limit on total stack usage)

+/- **Scope/lifetime dictated by control flow**

  Private to stack frame

  Does not persist after function exits

Why do we also need a heap?

An example:

`code/heap/names.c`

Dynamic storage

+ Programmer controls scope/lifetime

Versatile, precise

Works for situations where global/stack do not

- Needs software runtime support

Library module to manage heap memory

Functions to allocate/deallocate memory (explicit calls by client)

- C version is low on safety

No type safety (raw `void*`, size in number of bytes)

Much opportunity for error

(allocate wrong size, use after free, double free)

Heap module interface

```
void *malloc (size_t nbytes);
void  free (void *ptr);
```

What is a `void*` pointer?

"Generic" pointer, holds a memory address

Type of pointee is not specified, could be any type of data

What you can do with a `void*`

Pass to/from function, assignment

What you <u>cannot</u> do with a `void*`

No dereference without cast

No pointer arithmetic without cast (scaling unknown!)

No array indexing (size of pointee unknown!)

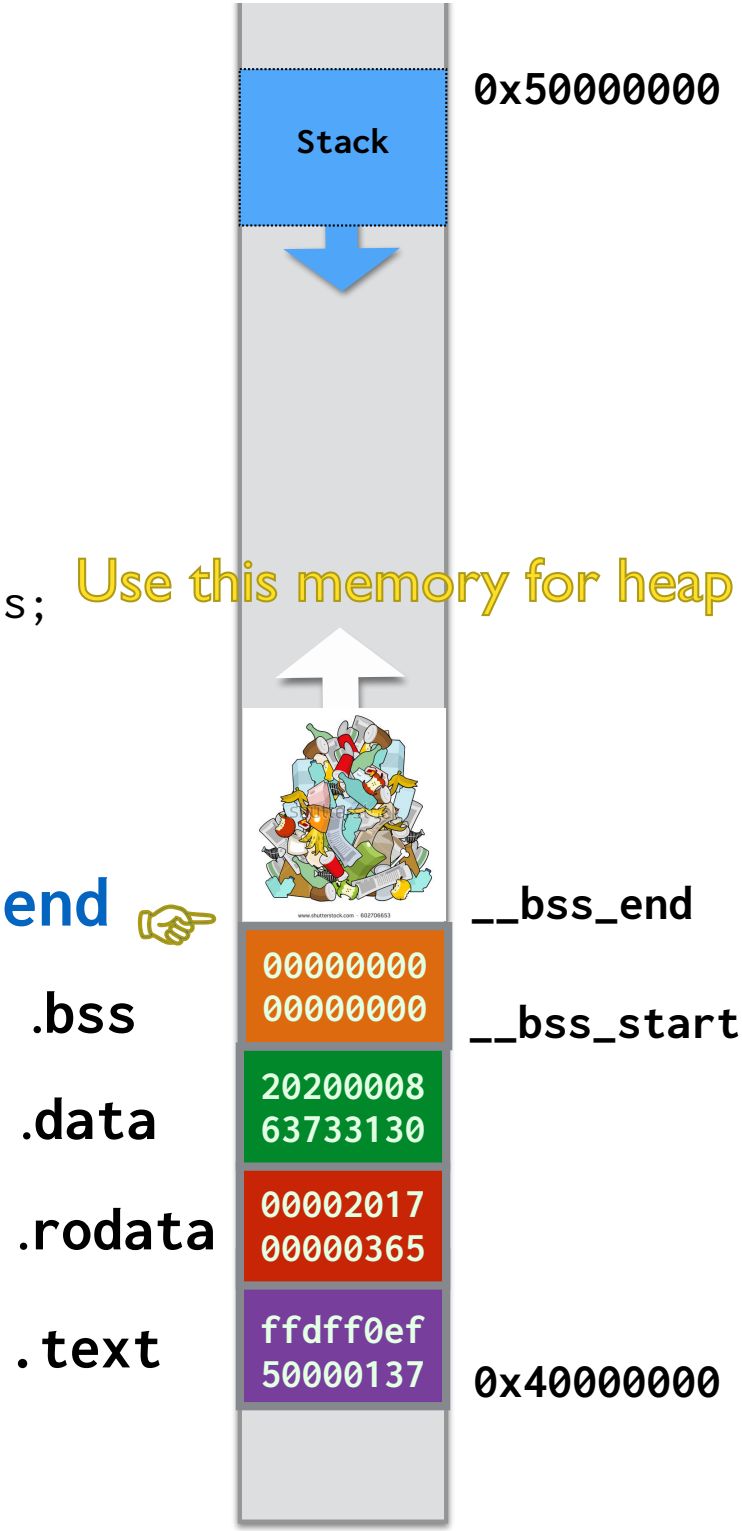
How to implement a heap

Drawing by Jane Lange

Bump allocator

```
void *sbrk(int nbytes) {
    static void *cur_heap_end = &__bss_end;

    void *prev_end = cur_heap_end;
    cur_heap_end = (char *)cur_heap_end + nbytes;
    return prev_end;
}
```

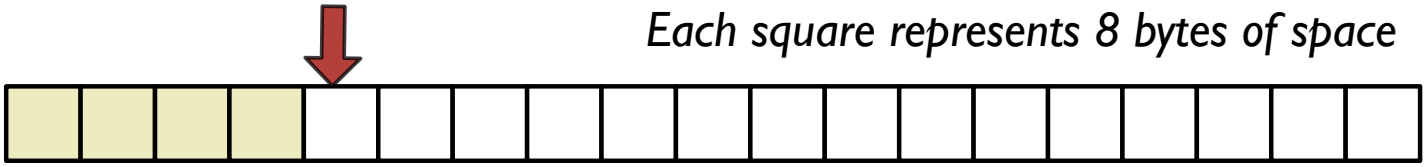


- Keep pointer to end of heap segment
- Service malloc request adjusts pointer upward
- Every request extends/grows heap segment
- No reuse/recycle

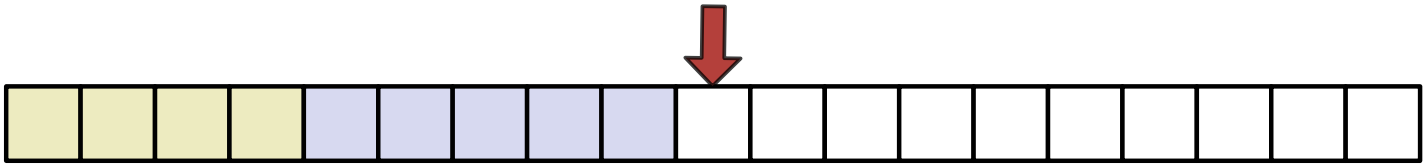
Tracing the bump allocator

Each square represents 8 bytes of space

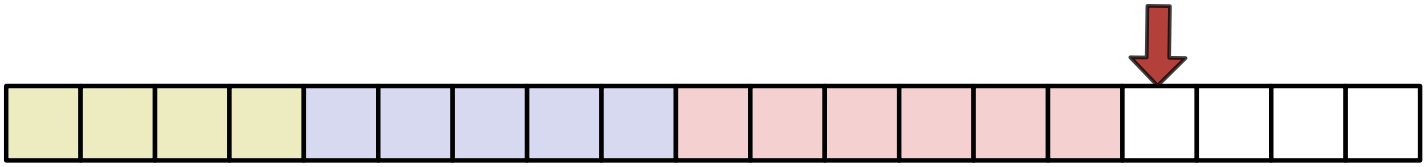
```
p1 = malloc(32)
```



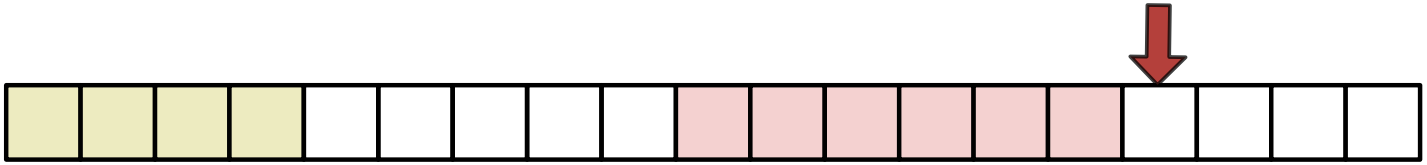
```
p2 = malloc(40)
```



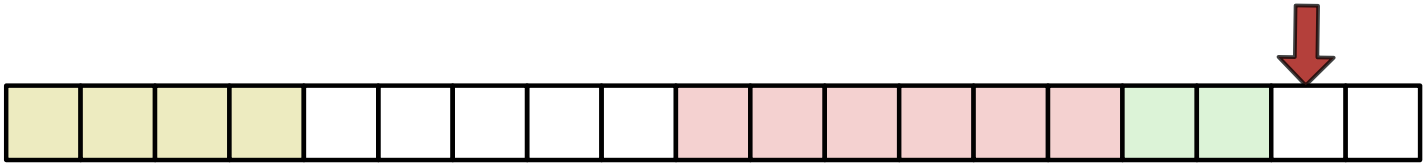
```
p3 = malloc(48)
```



```
free(p2)
```



```
p4 = malloc(16)
```



Bump Memory Allocator

`code/heap/malloc.c`

Evaluate bump allocator

+ Operations super-fast

+ Very simple code, easy to verify, test, debug

- No recycling/re-use

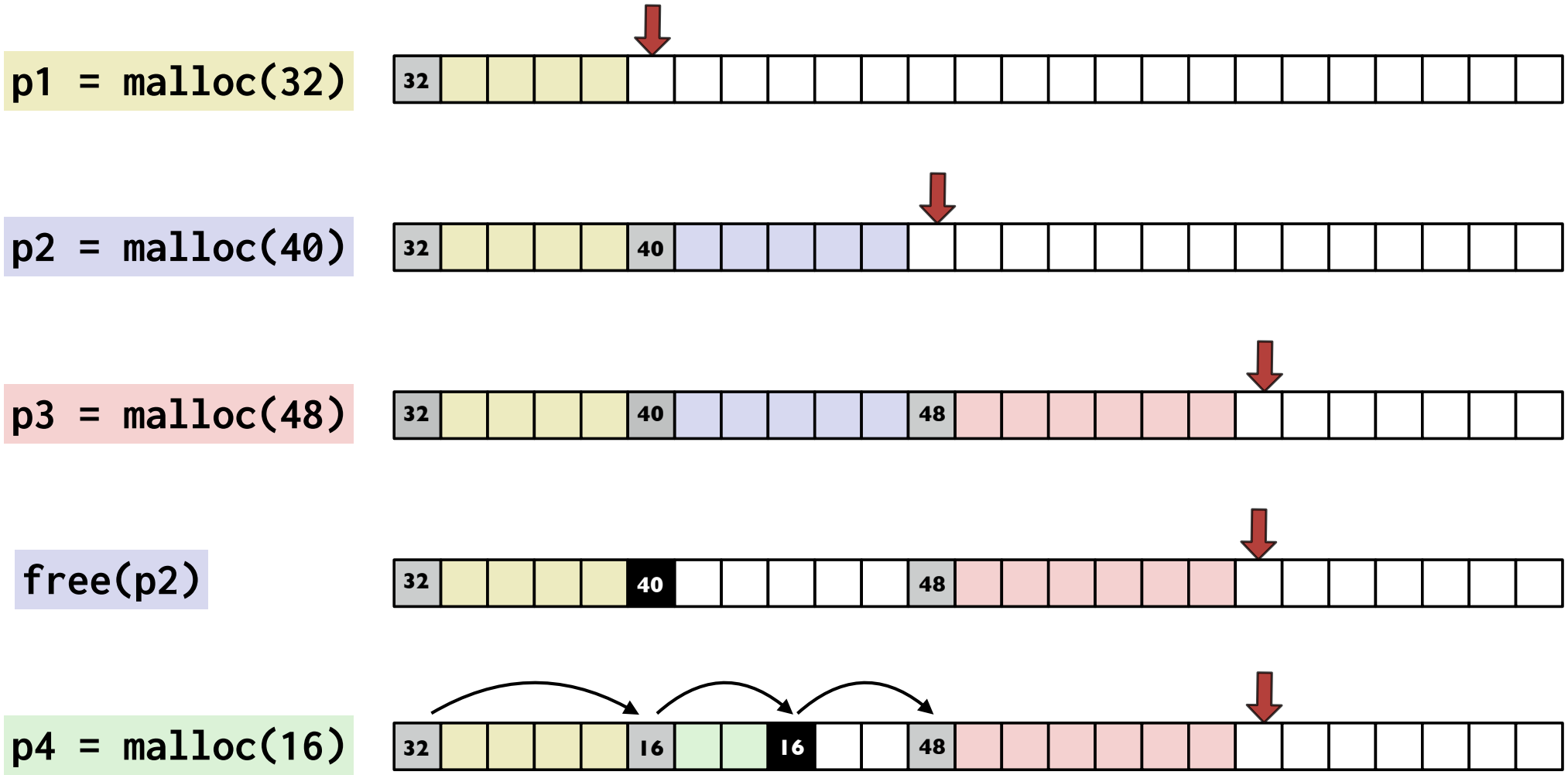
  (in what situations will this be problematic?)

- Sad consequences if **`sbrk()`** advances into stack

  (what can we do about that?)

Pre-block header, implicit list

Each square represents 8 bytes, header records size of payload in bytes



Header struct on each block

```
struct header {
    unsigned int size;
    unsigned int status;
};                              // sizeof(struct header) = 8 bytes

enum { IN_USE = 0, FREE = 1 };


void *malloc(size_t nbytes)
{
    nbytes = roundup(nbytes, 8);
    size_t total_bytes = nbytes + sizeof(struct header);

    struct header *hdr = sbrk(total_bytes); // extend end of heap
    hdr->size = nbytes;
    hdr->status = IN_USE;
    return hdr + 1;     // return address at start of payload
}
```

Challenge for malloc client

Correct allocation (size in bytes)

Correct access to block (within bounds, not freed)

Correct free (once and only once, at correct time)

What happens if you…

— forget to free a block after you are done using it?
— access a memory block after you freed it?
— free a block twice?
— free a pointer you didn't malloc?
— access outside the bounds of a heap-allocated block?

Challenge for malloc implementor

just `malloc` is easy 😎
`malloc` with `free` is hard 🤔
Efficient `malloc` with `free` ….Yikes! 🥵

Complex code (pointer math, typecasts)
Critical system component
  correctness is non-negotiable!
Thorough testing is essential

Survival strategies:
  draw lots of pictures
  printf (you've earned it!!)
  early tests on inputs small enough to trace by hand if need be
  build up to more complex tests